

3M Window Film

Private House at Rama III project

Jun 23, 2023

Benefit Applying Window Film



Solution for Energy Efficiency

Solar Control Films

Window films are a type of window treatment applied to windows and glass surfaces, designed to reduce the amount of solar heat transmitted through the glass and increase safety and security. The films help increase solar reflectivity, reject solar heat and reduce incoming ultraviolet light. 3M™ Window Films can significantly reduce the amount of heat coming through your windows to help lower utility costs.

3M Window Film can help you:

- Broad estimate electrical save 130-205 kwh/m2
- Reduce electricity bills as much as 10%
- Reduce temperatures as much as 9°F(5°C)
- Earn as many as 11 LEED points
- Increase your ENERGY STAR Portfolio Manager score
- Obtain utility rebates to help provide faster payback



LEED credit

- | | |
|---|---|
| • SS-8 (Light Pollution Reduction) | • MR 1.1-1.2 (Retain 75-95% of existing walls, floors and roof) |
| • EQ-7.1 (Thermal Comfort) | • EQ-8.1-8.2 (Daylight 75-95% of spaces) |
| • EA-1 (Optimized Energy Performance) | • MR 5.1-5.2 (Extracted, Processed & Manufactured Regionally) |
| • EQ-7.2 (Thermal Comfort Verification) | • ID (Environmental quality UV rejection) |

Solution for Energy Efficiency

Solar Control Films

Department of Energy (FEMP) Study Results

- **Maximum cost effectiveness rating**
- **Maximum probability of success**
- Window films are considered “top-tier technology”
- Window films have a typical **payback of:**

less than

3 years

CONSOL Energy Study Results

- Window films **pay back in <3 years**
- Window films **save up to:**

19 kWh/ft² on a **single pane**
or

12 kWh/ft² on a **double pane**

3M Solar Control Films have been proven to effectively improve energy savings

Department of Energy (DOE) Study

Background

In 2011, the US DOE completed a study on the top 50 commercially available energy conservation technologies. The technologies were ranked in these categories:

- Payback
- Probability of success
- Overall energy savings
- Technologies with less benefit that may be considered for specific targeted applications

Results

- Ranked as top-tier technology
- **Fastest payback ranking available — approximately 3 years**
- Highest probability of success ranking (based on customer acceptance, ease of retrofit, knowledge base of the technology and supply chain strength)

ConSol Study

Background

ConSol Energy and Environmental Solutions is a leading consulting firm for builders, government agencies, utilities and trade associations. ConSol utilized the US DOE-recommended software platform, Energy Plus, to calculate the effects of adding window film to a commercial building. The commercial building model used was the DOE recommended Energy Plus Commercial Building Benchmark Model. The study was completed in ICC Climate Zones 1, 2 and 3.

Results

This study further justified the DOE study with the following results:

- **Single-pane glass showed paybacks in as short as 1.4 years**
 - **Double-pane glass showed paybacks in as short as 2.1 years**
 - Annual energy savings as much as 19 kWh/sq ft (200 kWh/sq m) of glass
 - An 8.8% carbon emission reduction would occur if every home in California had window film installed
- After this study was published in 2011, the state of California updated its building code to include window films.²

Proven savings in every climate.

The US is broken down into eight different climate zones in accordance with building codes today — Climate Zone 1 being the hottest and Climate Zone 8 the coldest. 3M™ Window Films can save energy in all climate zones.

Climate Zone 1

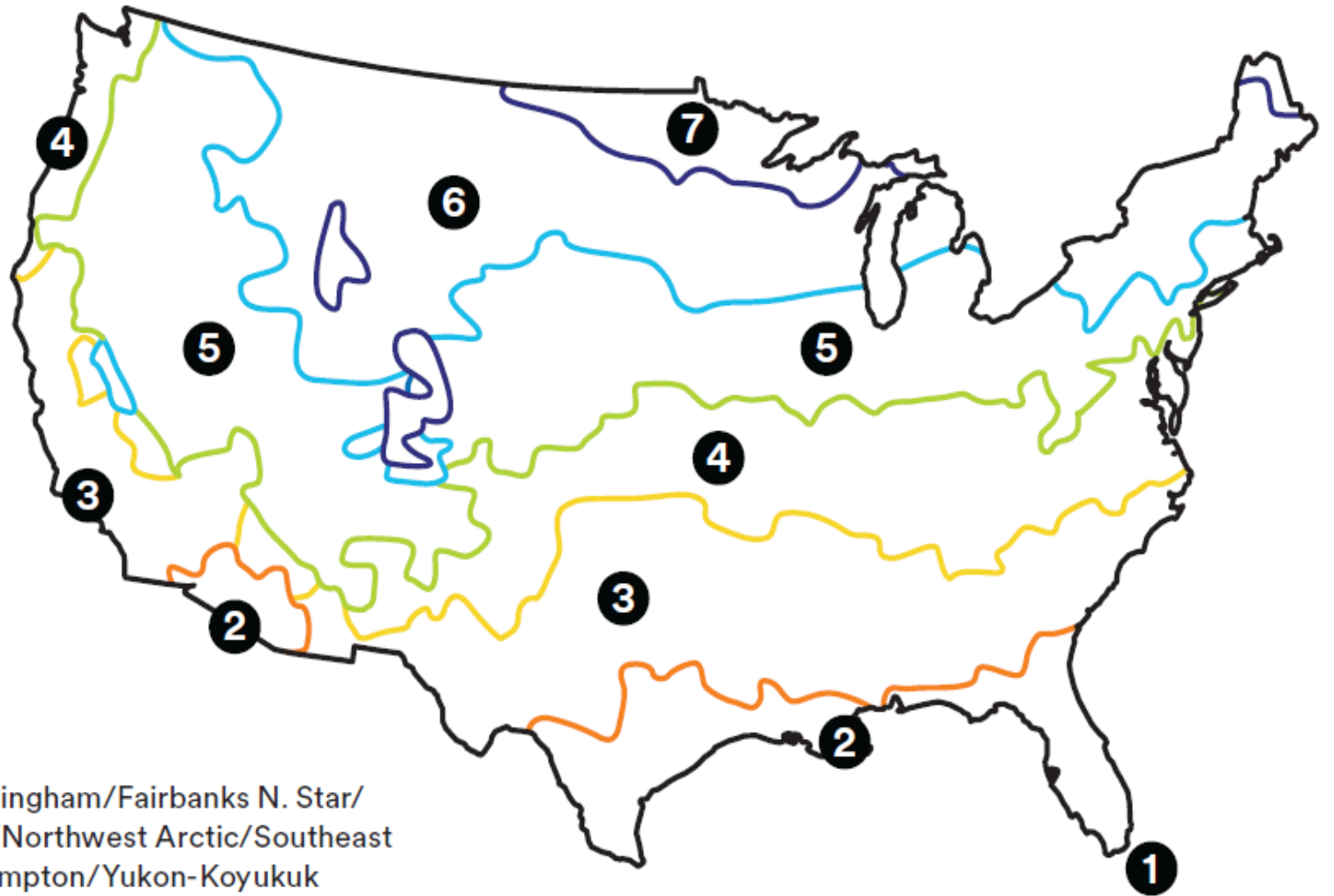
Includes: Guam/Hawaii/
Puerto Rico/Virgin Islands

Climate Zone 7

Includes: Alaska
Some Alaska boroughs are
also included in Zone 8

Climate Zone 8

Includes: Bethel/Dillingham/Fairbanks N. Star/
Nome/North Slope/Northwest Arctic/Southeast
Fairbanks/Wade Hampton/Yukon-Koyukuk





Climate Zone 1 | Case Study:

Mount Terrace Condominium, Hawaii Kai, Hawaii

Challenge: To retain a consistent look for the condominium complex while also finding ways to reduce the cost of managing the building.

Solution: 3M™ Sun Control Window Film Prestige 50 was chosen to obtain a maximum amount of energy rejection while maintaining the current look of the building. In addition, the 3M™ Prestige Series Window Film product was chosen because it has no metals and therefore no possibility of corrosion from the ocean air.

Results:

Payback — Less than 2 years

Savings — Greater than \$270,000 annually

Energy saved — 17 kWh/sq ft (180 kWh/sq m) of glass

Date completed — 2012



Climate Zone 2 | Case Study:

National Bank of Arizona, Phoenix, Arizona

Challenge: To retain the natural lighting while reducing both the heat it generated and glare on the banking floor. Minimizing disruption to normal banking operations was paramount.

Solution: 3M™ Sun Control Window Film Prestige Exterior 40 was installed because of its high infrared heat and harmful ultraviolet ray rejection. It is non-metallic and thus will not corrode, negating the need for edge sealing and minimizing installation time. Much of the project's cost was offset by an energy-efficiency rebate from Salt River Project (SRP).

Results:

Glass — Double pane tinted

Payback — Less than 5 years

Savings — Greater than \$25,000 annually

Energy saved — 12 kWh/sq ft (130 kWh/sq m) of glass

Date completed — 2011



Climate Zone 3 | Case Study:

Century Plaza Towers, Los Angeles, California

Challenge: To solve interior temperature imbalance and make the towers more energy efficient without drastically changing the look of the historic buildings.

Solution: 3M™ Sun Control Window Film Neutral 35, which offers high heat rejection and neutral light appearance, was installed. Much of the project's cost was offset by a rebate of \$116,000 from the Los Angeles Department of Water and Power.

Results:

Payback — Less than 1 year

Savings — Greater than \$200,000 annually

Energy Saved — Greater than 15 kWh/sq ft
(161 kWh/sq m) of glass

Date Completed — 2009



Climate Zone 4 | Case Study:

Fifth Third Center, Cincinnati, Ohio

Challenge: To control incoming light and heat in order to keep tenants comfortable, reduce energy costs and maintain the building's striking aesthetics.

Solution: Solar Tint, which had installed 3M™ Window Film in 1980, was chosen to update the facility's window film to further reduce energy costs by installing 3M™ Sun Control Window Film Neutral 20.

Results:

Payback — Less than 1 year

Savings — Greater than \$100,000 annually

Date Completed — 2000



Climate Zone 5 | Case Study:

1501 Clinton Avenue, Baltimore, Maryland

Challenge: To non-intrusively cut energy costs and increase interior comfort in a short time frame without affecting the tower's reflective modernism. Paramount was the issue of paying for up-front job costs on a limited budget.

Solution: 3M™ Sun Control Window Film Ceramic Series 30, which offers high clarity and outstanding heat reduction, was installed. To help pay for the project, 3M facilitated a custom rebate program with ICF International, the entity that supports Baltimore Gas and Electric's (BGE) energy efficiency programs.

Results:

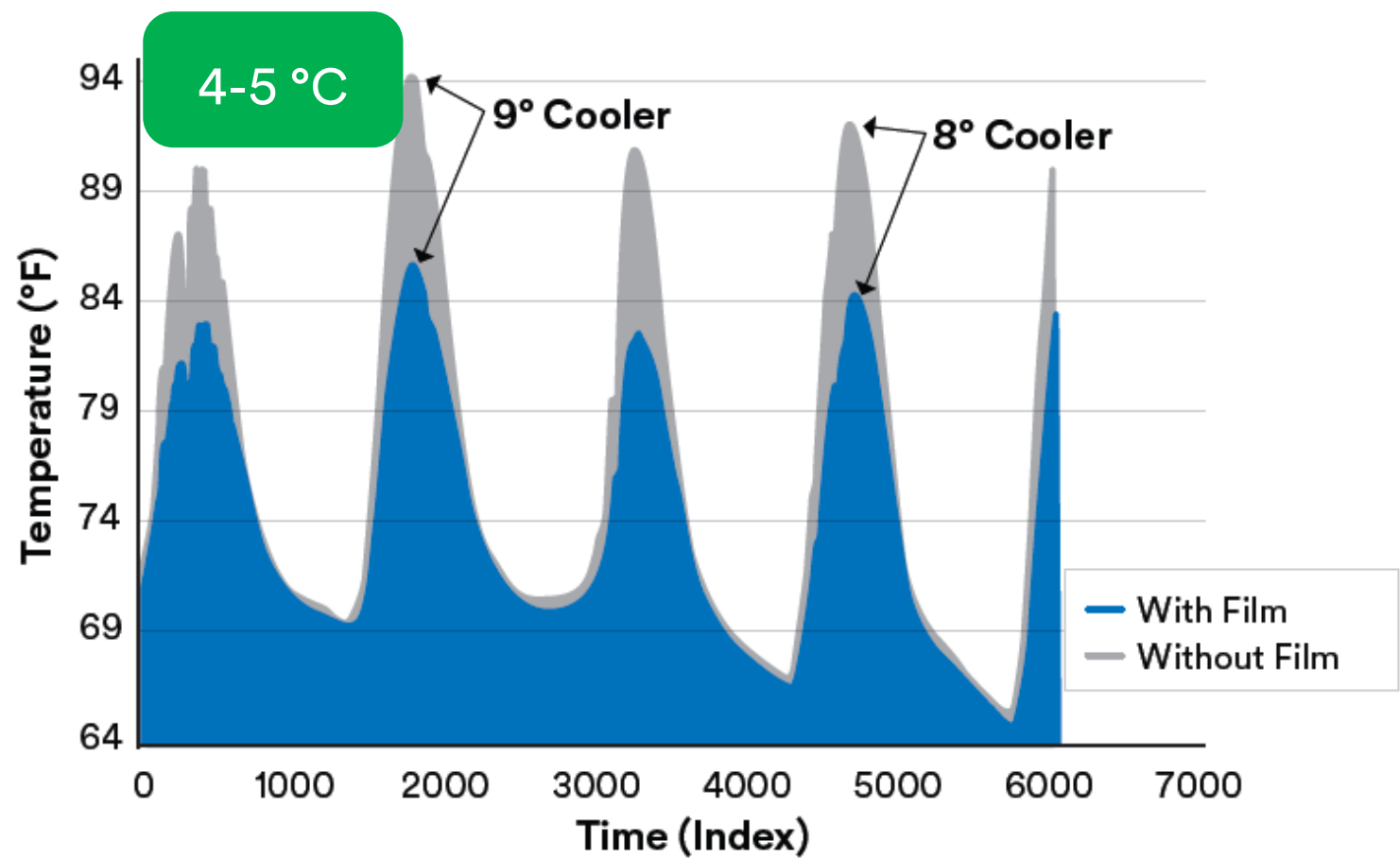
Payback — Less than 1 year

Savings — Greater than \$100,000 annually

Date completed — 2000

Comfortably cool.

3M™ Sun Control Window Film can reduce the temperature in direct sunlight by as much as 9° F, making your space much more usable and comfortable. The graph below shows a four-day temperature logging experiment conducted in an office building.

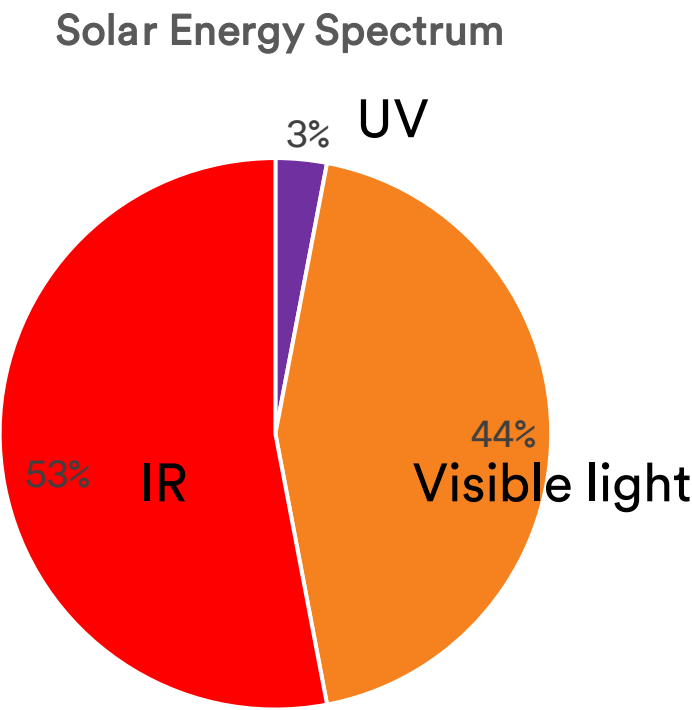
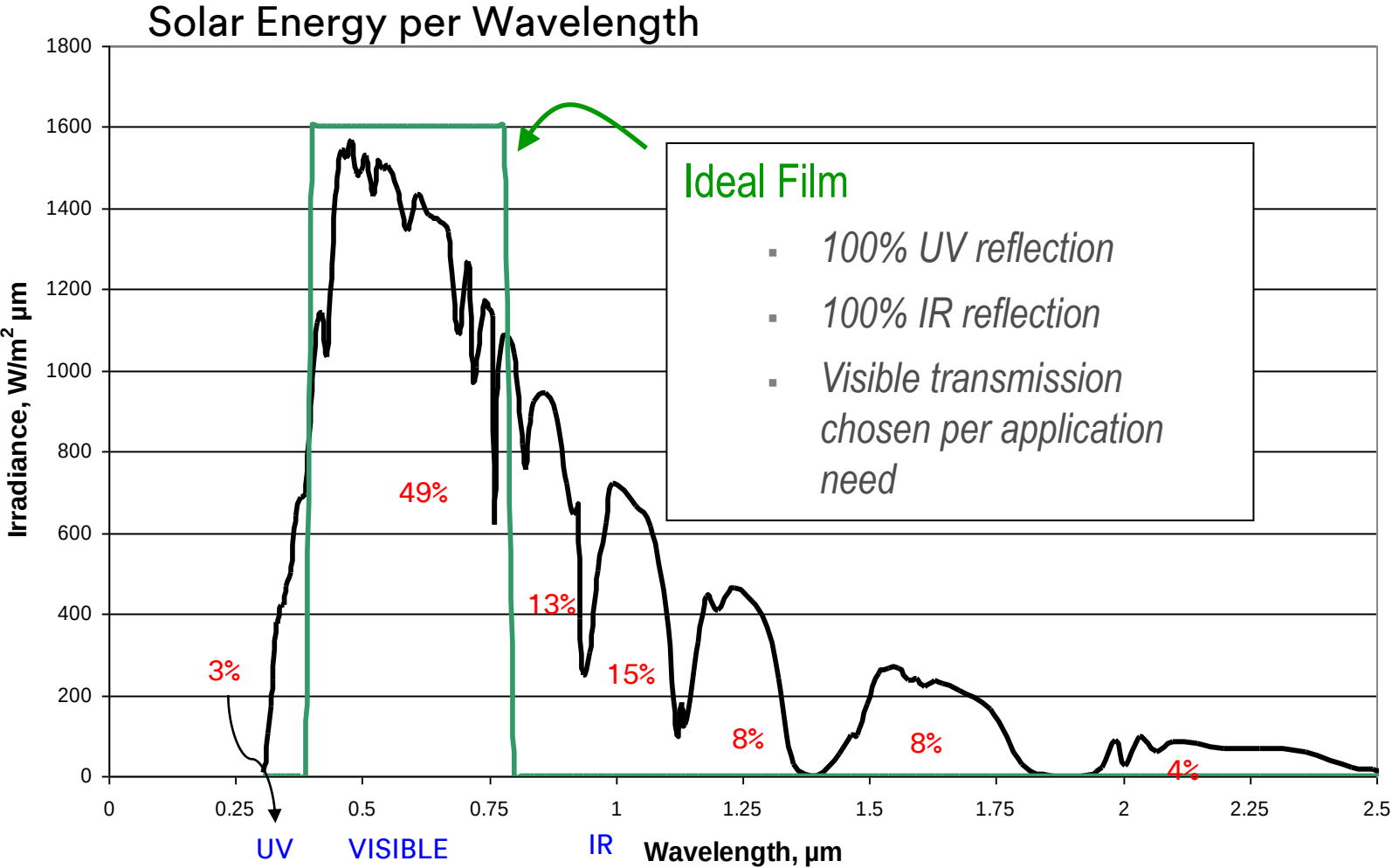


LEED Certification

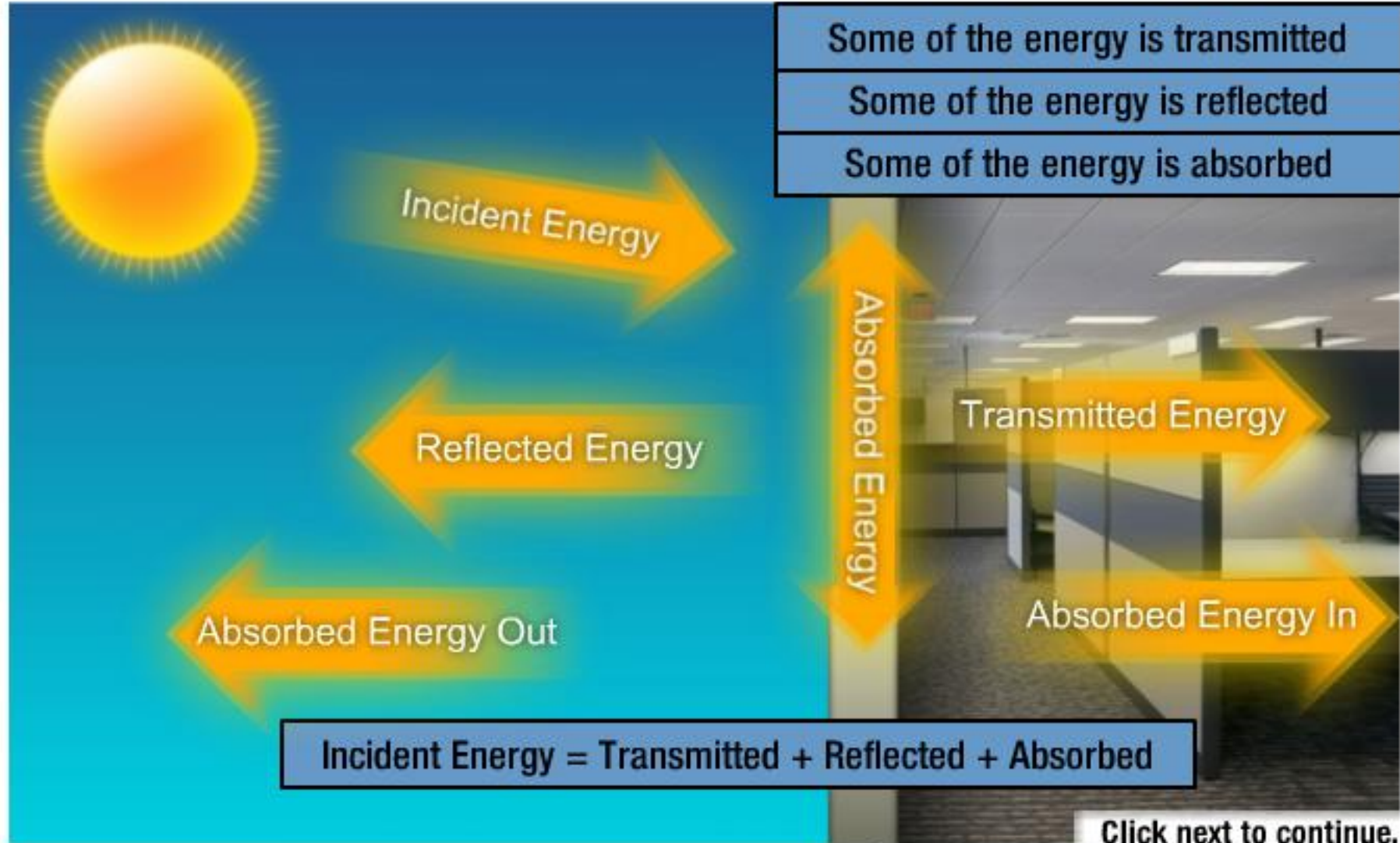
Window films may be used toward the following LEED credits:

▶ SS-8	▶ EQ-7.1
▶ EA-1	▶ EQ-7.2
▶ MR 1.1-1.2	▶ EQ-8.1-8.2
▶ MR 5.1-5.2	▶ ID

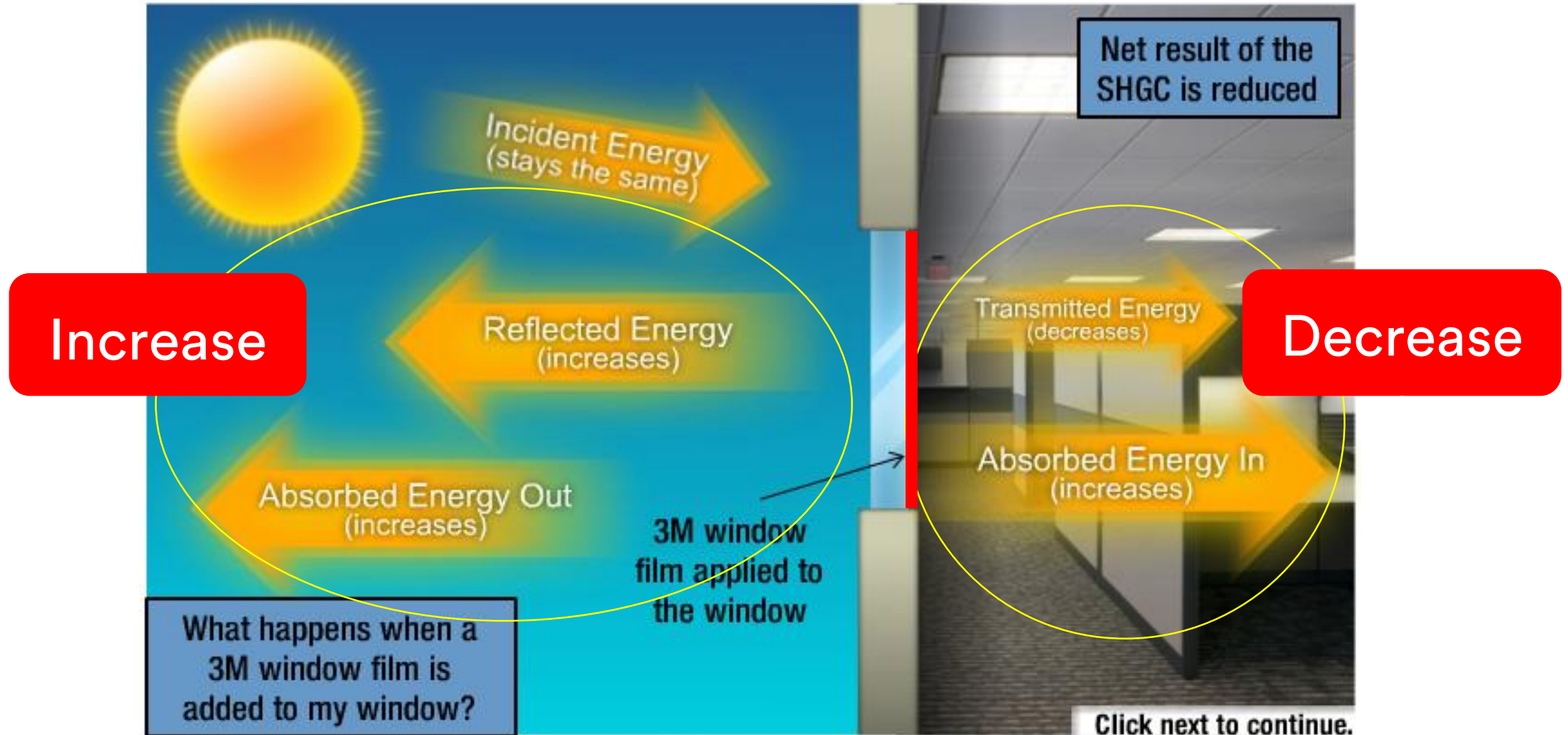
Window Film Basics



How Do Solar Films Work?

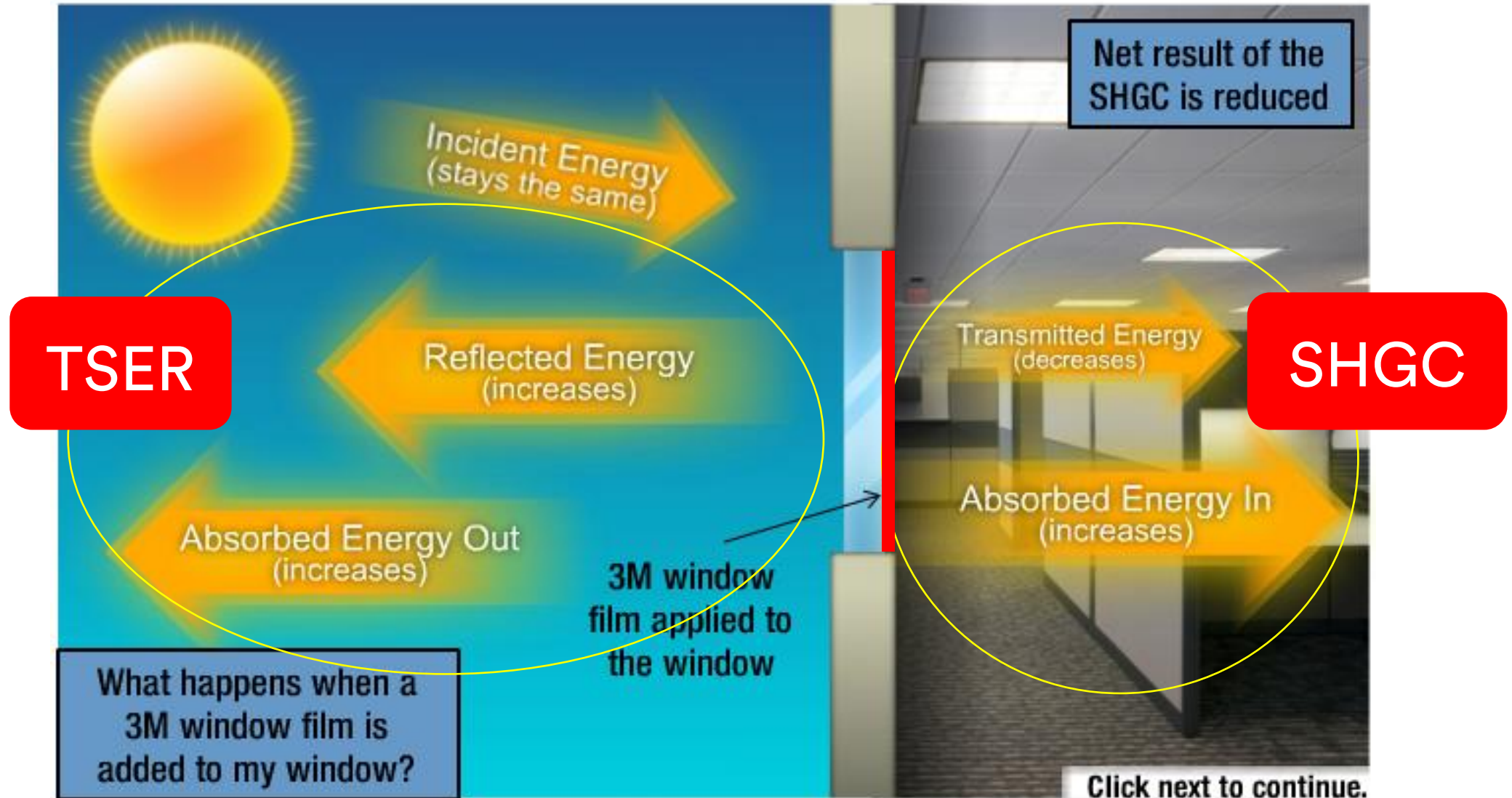


How Do 3M Solar Films Work?



3M After Installed Film

How Do 3M Solar Films Work?



3M After Installed Film

Benefit - Window Film

Can help :

1. Dramatic solar heat reduction.
2. Up to 99% UV block.
3. Better HVAC system management & lower air conditioning cost by reducing heat that comes into a building.
4. Save electricity cost.



Test result : Private House at Rama III road

Test window film

Objective of the Test

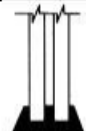
To test the performance of 3M Solar Control films 3M PR50 vs. No film by using the solar power meter.

Scope of the Test

1. Test Duration: This test was conducted on Jun23, 2023
2. Test Position: This test was conducted inside building. Panes of glass windows were selected to conduct the experiment in order to compare the difference of solar heat transfer through glass windows with 3M PR50 vs. No film.

Product specification: 3M Prestige 50

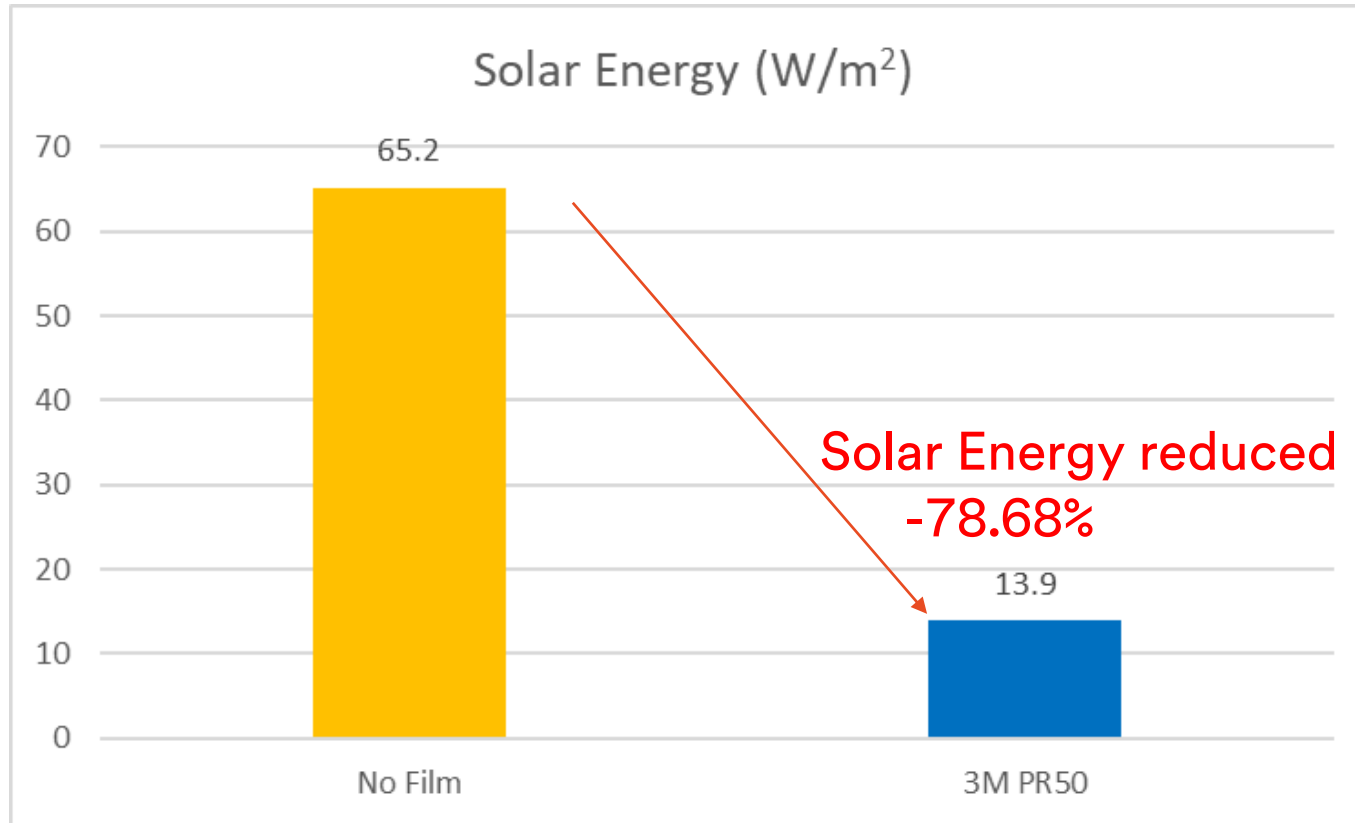
Product Performance & Technical Data

<u>PR50</u>								
	Single Pane		Tinted		Double Pane		Double tinted	
Film	1/4" Clear	PR50	1/4" tint	PR50	Dual 1/4" Clear	PR50	Dual 1/4" tint	PR50
Solar Heat Gain Coefficient	0.82	0.44	0.63	0.39	0.70	0.53	0.51	0.42
Visible Light Transmitted	88%	51%	53%	30%	79%	45%	47%	27%
Visible Light Reflected Interior	8%	7%	6%	6%	15%	9%	13%	9%
Visible Light Reflected Exterior	8%	8%	6%	6%	15%	14%	8%	8%
U Value	1.03	0.99	1.03	0.99	0.47	0.47	0.47	0.47
UV Block	43%	99.9%	NA	99.9%	NA	99.9%	NA	99.9%
Total Solar Energy Rejected	18%	56%	37%	61%	30%	47%	49%	58%
Glare Reduction	NA	44%	NA	43%	NA	43%	NA	44%
Heat Loss Reduction	NA	3%	NA	3%	NA	2%	NA	2%
Solar Heat Reduction	NA	46%	NA	38%	NA	24%	NA	18%

Testing location



1) Solar Energy Measurement



Solar Energy measured by Solar Power meter



No Film



3M PR50

The background is a complex, abstract geometric pattern composed of numerous overlapping triangles. The color palette is warm, featuring various shades of orange, from light yellow-orange to deep, rich orange and reddish-orange. The triangles vary in size and orientation, creating a dynamic and textured visual effect.

Thank you